

1.0 Elasticsearch installation and kibana

Step 1: Import the Elasticsearch PGP key

```
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --  
dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
```

Install from the APT repository.

1. install the `apt-transport-https` package on Debian

```
sudo apt-get install apt-transport-https
```

2. Save the repository definition to `/etc/apt/sources.list.d/elastic-9.x.list`:

```
echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg]  
https://artifacts.elastic.co/packages/9.x/apt stable main" | sudo tee  
/etc/apt/sources.list.d/elastic-9.x.list
```

3. Install the Elasticsearch Debian package:

```
sudo apt-get update && sudo apt-get install elasticsearch
```

```

depi@SOC:~/DEP1$ sudo apt-get update && sudo apt-get install elasticsearch
Hit:1 http://eg.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://eg.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://eg.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:5 https://artifacts.elastic.co/packages/9.x/apt stable InRelease [3,248 B]
Get:6 https://artifacts.elastic.co/packages/9.x/apt stable/main amd64 Packages [17.2 kB]
Fetched 20.5 kB in 1s (19.7 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  elasticsearch
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 676 MB of archives.
After this operation, 1,299 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/9.x/apt stable/main amd64 elasticsearch amd64 9.1.5 [676 MB]
Fetched 676 MB in 5min 6s (2,212 kB/s)
Selecting previously unselected package elasticsearch.
(Reading database ... 203210 files and directories currently installed.)
Preparing to unpack .../elasticsearch_9.1.5_amd64.deb ...
Creating elasticsearch group... OK
Creating elasticsearch user... OK
Unpacking elasticsearch (9.1.5) ...
Setting up elasticsearch (9.1.5) ...
----- Security autoconfiguration information -----

Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.

The generated password for the elastic built-in superuser is : h*2PDFAw8pdSAQwgELCI

If this node should join an existing cluster, you can reconfigure this with
'/usr/share/elasticsearch/bin/elasticsearch-reconfigure-node --enrollment-token <token-here>'
after creating an enrollment token on your existing cluster.

You can complete the following actions at any time:

Reset the password of the elastic built-in superuser with
'/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic'.

Generate an enrollment token for Kibana instances with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana'.

Generate an enrollment token for Elasticsearch nodes with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s node'.

```

Reset the password of the elastic built-in superuser with

```
/usr/share/elasticsearch/bin/elasticsearch-reset-password -i -u elastic
```

Generate an enrollment token for Kibana instances with

```
/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana
```

```
/usr/share/elasticsearch/bin/elasticsearch-reset-password -i -u kibana_system
```

```
# make the password for kibana_system is >> passowrd # same as in
kibana.yml file (show make them the same )
```

Test elastic if it operate

```
curl --cacert /etc/elasticsearch/certs/http_ca.crt -u elastic
https://localhost:9200
```

```
root@SOC:/usr/share/elasticsearch/bin# curl --cacert /etc/elasticsearch/certs/http_ca.crt -u elastic https://localhost:9200
Enter host password for user 'elastic':
{
  "name" : "SOC",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "nLZ8nonWTS0pc1D90-HIBA",
  "version" : {
    "number" : "9.1.5",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "90ee222e7e0136dd8ddb34015538f3a00c129b7",
    "build_date" : "2025-10-02T22:07:12.966975992Z",
    "build_snapshot" : false,
    "lucene_version" : "10.2.2",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

edit the elasticsearch.yml file

```
path >> '/etc/elasticsearch/elasticsearch.yml'
```

- 1- modify the network.host
- 2- uncomment the http.port

```
GNU nano 6.2                               elasticsearch.yml *
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ----- Network -----
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
network.host: 192.168.159.132
#
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
http.port: 9200
#
# For more information, consult the network module documentation.
#
# ----- Discovery -----
```

allow traffic on port 9200 tcp/udp to make elasticsearch port reachable

```
sudo ufw allow 9200/tcp
sudo ufw allow 9200/udp
```

```
root@SOC:/etc/elasticsearch# ufw allow 9200/tcp
Rules updated
Rules updated (v6)
root@SOC:/etc/elasticsearch# ufw allow 9200/udp
Rules updated
Rules updated (v6)
```

Kibana

```
sudo apt-get install kibana

sudo systemctl start kibana.service

sudo systemctl enable kibana.service
```

edit the kibana.yml file

```
path >> '/etc/kibana/kibana.yml'
```

```
GNU nano 6.2 kibana.yml
# Specifies the address to which the Kibana server will bind. IP addresses and host names are both valid values.
# The default is 'localhost', which usually means remote machines will not be able to connect.
# To allow connections from remote users, set this parameter to a non-loopback address.
server.host: "192.168.159.132"

# Enables you to specify a path to mount Kibana at if you are running behind a proxy.
# Use the `server.rewriteBasePath` setting to tell Kibana if it should remove the basePath
# from requests it receives, and to prevent a deprecation warning at startup.
# This setting cannot end in a slash.
#server.basePath: ""

# Specifies whether Kibana should rewrite requests that are prefixed with
# `server.basePath` or require that they are rewritten by your reverse proxy.
# Defaults to `false`.
#server.rewriteBasePath: false

# Specifies the public URL at which Kibana is available for end users. If
# `server.basePath` is configured this URL should end with the same basePath.
#server.publicBaseUrl: ""

# The maximum payload size in bytes for incoming server requests.
#server.maxPayload: 1048576

# The Kibana server's name. This is used for display purposes.
#server.name: "your-hostname"

# ===== System: Kibana Server (Optional) =====
# Enables SSL and paths to the PEM-format SSL certificate and SSL key files, respectively.
# These settings enable SSL for outgoing requests from the Kibana server to the browser.
#server.ssl.enabled: false
#server.ssl.certificate: /path/to/your/server.crt
#server.ssl.key: /path/to/your/server.key

# ===== System: Elasticsearch =====
# The URLs of the Elasticsearch instances to use for all your queries.
elasticsearch.hosts: ["https://192.168.159.132:9200"]

# If your Elasticsearch is protected with basic authentication, these settings provide
# the username and password that the Kibana server uses to perform maintenance on the Kibana
# index at startup. Your Kibana users still need to authenticate with Elasticsearch, which
# is proxied through the Kibana server.
elasticsearch.username: "kibana_system"
elasticsearch.password: "password"

# Kibana can also authenticate to Elasticsearch via "service account tokens".
# Service account tokens are Bearer style tokens that replace the traditional username/password based configuration.
```

```
GNU nano 6.2 kibana.yml
# headers, set this value to [] (an empty list).
#elasticsearch.requestHeadersWhitelist: [ authorization ]

# Header names and values that are sent to Elasticsearch. Any custom headers cannot be overwritten
# by client-side headers, regardless of the elasticsearch.requestHeadersWhitelist configuration.
#elasticsearch.customHeaders: {}

# Time in milliseconds for Elasticsearch to wait for responses from shards. Set to 0 to disable.
#elasticsearch.shardTimeout: 30000

# ===== System: Elasticsearch (Optional) =====
# These files are used to verify the identity of Kibana to Elasticsearch and are required when
# xpack.security.http.ssl.client_authentication in Elasticsearch is set to required.
#elasticsearch.ssl.certificate: /path/to/your/client.crt
#elasticsearch.ssl.key: /path/to/your/client.key

# Enables you to specify a path to the PEM file for the certificate
# authority for your Elasticsearch instance.
elasticsearch.ssl.certificateAuthorities: [ "/etc/kibana/certs/http_ca.crt" ]

# To disregard the validity of SSL certificates, change this setting's value to 'none'.
#elasticsearch.ssl.verificationMode: full
```

Generate 3 random Encryption keys

1. `xpack.encryptedSavedObjects.encryptionKey`

Used to encrypt saved objects like dashboards, searches, and visualizations.

2. `xpack.security.encryptionKey`

Used to secure user sessions and authentication features.

3. `xpack.reporting.encryptionKey`

Used to encrypt reports (PDF, CSV, exports) during storage and transport.

- i generate this keys using python by secret library

```
1 import secrets
2 print(secrets.token_urlsafe(32))
3 print(secrets.token_urlsafe(32))
4 print(secrets.token_urlsafe(32))
5
```

in  generate secret keys x

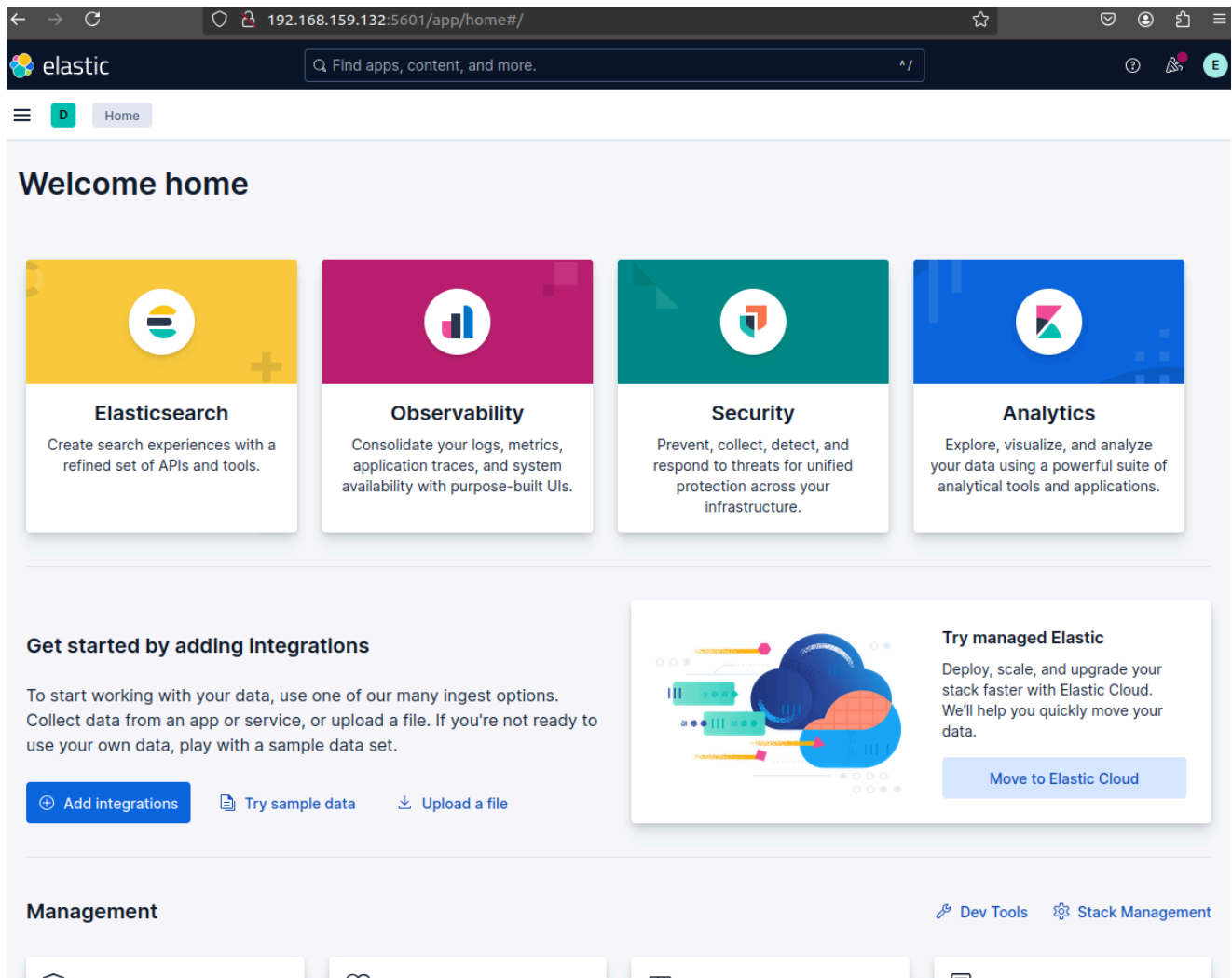
```
C:\Users\terbo\PycharmProjects\pythonProject3\venv\
fvhgBrxgOdBBG3UH4JqnbSW4JizYfeH6h249e8-LOU0
gUSCoR6QuRa8vjW6rSodSCeT2mS7FXqK8CYwQZgWX00
cIVEqBfGPbrpW-gzeWz5rpK3zuUcIFbZbw0HxH40868
```

- then add this three keys in **kibana** configuration file `kibana.yml`

```
#kibana.search.autoComplete.valuesuggestions.tokenizer: "lucene"
xpack.encryptedSavedObjects.encryptionKey: "9832219bc3413bc85960d8c83d6c83a74339a5eb2e4e1bc5c70bfb0fa0da3a1e"
xpack.security.encryptionKey: "fcbac5f4414558e3e40764e9ac1a63a7226cffaa6282720aa48ef29ece55b5c9"
xpack.reporting.encryptionKey: "4f37399dd120788c9ada5e349b6872ea71e99e904c0718f23401f1c7da65b6d6"
```

```
cd /etc/kibana
mkdir certs
cp /etc/elasticsearch/certs/http_ca.crt /etc/kibana/certs/ # this is
certiface authority certifation issued from elasticsearch to enabel https
sudo systemctl restart kibana.service
```

elasticsearch account
elastic:Hunter@2025



know the integration between elasticsearch and kibana success .

Install Fleet server

importance of installing a Fleet Server:

1. **Centralized Management:** Fleet Server acts as a central hub, enabling streamlined management and monitoring of all agents across your environment.
2. **Efficient Data Collection:** It ensures reliable collection, forwarding, and indexing of logs, metrics, and security events to your SIEM or analytics platform.
3. **Enhanced Security & Visibility:** Installing Fleet Server improves real-time threat detection, visibility, and control over endpoints and networked systems.

Step : 1

Fleet
Centralized management for Elastic Agents.

Agents Agent policies Enrollment tokens Uninstall tokens Data streams

Add a Fleet Server

A Fleet Server is required before you can enroll agents with Fleet. Follow the instructions below to set up a Fleet Server. For more information, see the [Fleet and Elastic Agent Guide](#).

Quick Start Advanced

- 1 Get started with Fleet Server**

First, set the public IP or host name and port that agents will use to reach Fleet Server. It uses port **8220** by default. We'll then generate a policy for you automatically.

Name
fleet-depi-server

URL
https://192.168.159.132:8220

+ Add another URL

Generate Fleet Server policy
- 2 Install Fleet Server to a centralized host**
- 3 Confirm connection**

Step : 2

- choose the architecture type of your machine
- you can use `uname -m` to know it (i use **ubuntu 22** so it i choose >> Linux x86_64)

Add a Fleet Server

A Fleet Server is required before you can enroll agents with Fleet. Follow the instructions below to set up a Fleet Server. For more information, see the [Fleet and Elastic Agent Guide](#) .

Quick Start

Advanced

Install Fleet Server agent on a centralized host so that other hosts you wish to monitor can connect to it. In production, we recommend using one or more dedicated hosts. For additional guidance, see our [installation docs](#) .

Linux aarc...

MacOS aarc...

DEB aarc...

RPM aarc...

1 ▾

```
curl -L -O https://artifacts.elastic.co/downloads/elastic-agent-9.1.5-linux-x86_64.tar.gz
tar xzvf elastic-agent-9.1.5-linux-x86_64.tar.gz
cd elastic-agent-9.1.5-linux-x86_64
sudo ./elastic-agent install \
  --fleet-server-es=http://localhost:9200 \
  --fleet-server-service-token=AAEAAWVsYXN0aWMvZmxlZXQtc2VydmVyL3Rva2VuLTE3NjA5ODU0ODk3NDM6YUducE8zdnltCg== \
  --fleet-server-policy=fleet-server-policy \
  --fleet-server-port=8220 \
  --install-servers
```

Windows x86_64

Windows MSI

✓ [Linux x86_64](#)

MacOS x86_64

DEB x86_64

RPM x86_64

3

Confirm connection

```
sudo ./elastic-agent install \
  --fleet-server-es=https://192.168.159.132:9200 \
  --fleet-server-es-insecure \      ## add this in terminal manually to bypass C
  --fleet-server-service-
token=AAEAAWVsYXN0aWMvZmxlZXQtc2VydmVyL3Rva2VuLTE3NjA5ODU0ODk3NDM6YUducE8zdnltCg== \
  --fleet-server-policy=fleet-server-policy \
  --fleet-server-port=8220 \
  --install-servers --insecure
```

```

depi@SOC:~/new$ sudo tar xzvf elastic-agent-9.1.5-linux-x86_64.tar.gz
cd elastic-agent-9.1.5-linux-x86_64
sudo ./elastic-agent install \
  --fleet-server-es=http://192.168.159.132:9200 \
  --fleet-server-service-token=AAEAAWVsYXN0aWMvZmxlZXQtc2VydMvYl3Rva2VuLTE3NjE3NTUyNTU0NDU6VmpwRDhod1RTNlNjSVhia3FOanNNQQ \
  --fleet-server-policy=fleet-server-policy \
  --fleet-server-port=8220 \
  --install-servers --insecure
[sudo] password for depi:
elastic-agent-9.1.5-linux-x86_64/.elastic-agent.active.commit
elastic-agent-9.1.5-linux-x86_64/data/elastic-agent-7bee01/otelcol
elastic-agent-9.1.5-linux-x86_64/data/elastic-agent-7bee01/elastic-agent
elastic-agent-9.1.5-linux-x86_64/data/elastic-agent-7bee01/package.version
elastic-agent-9.1.5-linux-x86_64/manifest.yaml
elastic-agent-9.1.5-linux-x86_64/.build_hash.txt
elastic-agent-9.1.5-linux-x86_64/LICENSE.txt
elastic-agent-9.1.5-linux-x86_64/otel_samples/
elastic-agent-9.1.5-linux-x86_64/otel_samples/autoops_es.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/gateway.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/logs_metrics_traces.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/managed_otlp/
elastic-agent-9.1.5-linux-x86_64/otel_samples/managed_otlp/logs_metrics_traces.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/managed_otlp/platformlogs.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/managed_otlp/platformlogs_hostmetrics.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/platformlogs.yml
elastic-agent-9.1.5-linux-x86_64/otel_samples/platformlogs_hostmetrics.yml
elastic-agent-9.1.5-linux-x86_64/otelcol
elastic-agent-9.1.5-linux-x86_64/NOTICE.txt
elastic-agent-9.1.5-linux-x86_64/elastic-agent.reference.yml

```

Fleet
Agents
Send feedback

Fleet

Centralized management for Elastic Agents.

[Agents](#)
[Agent policies](#)
[Enrollment tokens](#)
[Uninstall tokens](#)
[Data streams](#)
[Settings](#)

Ingest Overview Metrics
Agent Info Metrics
Agent activity
Add Fleet Server
Add agent

Status 5
Tags 1
Agent policy 4
Upgrade available

Showing 2 agents
Clear filters
Healthy 1
Unhealthy 0
Orphaned 0
Updating 0
Offline 1
Inactive 0
Unenrolled 0
Uninstalled 0

☐	Status	Host	Agent policy	CPU	Memory	Last activity	Version	Actions
☐	Healthy	SOC	Fleet Server Policy rev. 10	1.77 %	498 MB	31 seconds ago	9.1.5 ↑ Upgrade available	...

Not Securehttp://192.168.159.132:5601/app/Fleet/settings

☆

Sign in

elastic

Find apps, content, and more.

D

Fleet

Settings

Send feedback

Fleet

Centralized management for Elastic Agents.

AgentsAgent policiesEnrollment tokensUninstall tokensData streamsSettings

Fleet server hosts

Specify the URLs that your agents will use to connect to a Fleet Server. If multiple URLs exist, Fleet will show the first provided URL for enrollment purposes. For more information, see the [Fleet and Elastic Agent Guide](#).

Name	Host URLs	Default	Actions
fleet-depi-server	https://192.168.159.132:8220	✓	✎

[+ Add Fleet Server](#)

Outputs

Specify where agents will send data. [↗](#)

Name	Type	Hosts	Status	Default	Actions
default	Elasticsearch	http://192.168.159.132:9200		Agent integrationsAgent monitoring	✎

[+ Add output](#)

Agent Binary Download

List of sources used by Elastic agents to download their binaries. The default source applies to all agent policies that don't have a specific binary source specified.

Name	Host	Default	Actions
Elastic Artifacts	https://artifacts.elastic.co/downloads/	✓	✎